

Tax Payout Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Tax Payout Burden (TPB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on TPB achieves an annualized gross (net) Sharpe ratio of 0.28 (0.25), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 33 (28) bps/month with a t-statistic of 2.92 (2.49), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Change in equity to assets, Long-term EPS forecast, Payout Yield, Asset growth, Change in current operating assets) is 35 bps/month with a t-statistic of 2.84.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, with cross-sectional return differences arising from differential exposures to states where marginal utility is high. While the consumption-based asset pricing paradigm provides a coherent framework for understanding risk premia, identifying firm characteristics that capture consumption risk exposure remains a central challenge in financial economics. Tax payout burden—the ratio of tax payments to total distributions—emerges as a potentially important signal of firms’ sensitivity to aggregate consumption dynamics. Firms with high tax payout burdens face constrained flexibility in their capital allocation decisions during economic downturns, when both corporate tax obligations remain rigid and investor marginal utility peaks. This inflexibility in adjusting distributions when consumption growth slows suggests that tax payout burden may capture an important dimension of consumption risk exposure not fully reflected in existing factor models.

The tax payout burden signal connects to consumption risk through firms’ differential abilities to smooth cash flows across states of the world. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), assets that covary more strongly with consumption growth command higher risk premia because they fail to provide hedging benefits when marginal utility is high. Firms with elevated tax payout burdens face binding constraints on their distribution policies precisely when aggregate consumption declines and disaster risk increases [Barro \(2006\)](#). The rigidity of tax obligations relative to discretionary payouts creates state-dependent exposure to consumption shocks—high tax burden firms cannot adjust their capital structure efficiently when the stochastic discount factor is high, amplifying their sensitivity to systematic consumption risk.

The mechanism operates through the intertemporal marginal rate of substitution channel emphasized in [Campbell and Cochrane \(1999\)](#). During bad economic states

characterized by low consumption growth or heightened uncertainty, investors with habit formation preferences demand higher compensation for holding firms whose cash flow constraints prevent optimal consumption smoothing. Tax payout burden captures this state dependence because tax obligations represent non-discretionary claims that intensify during periods of already-elevated marginal utility. Unlike voluntary distributions that firms can adjust to buffer consumption shocks, tax payments create procyclical pressure on firm resources when the representative agent’s risk aversion peaks [Constantinides \(1990\)](#).

The predictive power of tax payout burden for returns should manifest most strongly through its correlation with consumption-based factors rather than purely financial metrics. In the framework of [Parker and Julliard \(2005\)](#), firms with high tax burdens exhibit greater sensitivity to aggregate consumption innovations because their inflexible payout structures prevent the precautionary savings behavior that would otherwise dampen consumption risk exposure. This creates a direct link between tax payout burden and the pricing kernel—high tax burden firms load more heavily on adverse consumption states, generating the positive return premium we observe empirically. The signal thus captures a fundamental aspect of how firms’ operational constraints translate into systematic risk exposure through the consumption channel.

We find that a value-weighted long/short trading strategy based on TPB achieves an annualized gross Sharpe ratio of 0.28, with the strategy earning average monthly returns of 28 basis points (t -statistic = 2.09). The strategy’s alpha relative to the [Fama and French \(2018\)](#) six-factor model equals 33 basis points per month with a t -statistic of 2.92, demonstrating robust abnormal returns after controlling for market, size, value, profitability, investment, and momentum factors. The long/short portfolio exhibits a significant negative loading of -0.52 (t -statistic = -6.78) on the CMA factor, consistent with tax-constrained firms facing higher costs of adjusting

investment during periods of elevated marginal utility.

The TPB signal’s predictive power persists across alternative portfolio construction methodologies and after accounting for transaction costs. Net of trading costs, the value-weighted quintile strategy using NYSE breakpoints generates monthly returns of 25 basis points (t-statistic = 1.90) with a net six-factor alpha of 28 basis points per month (t-statistic = 2.49). Among the largest stocks—those above the 80th NYSE percentile—the TPB strategy achieves average returns of 25 basis points per month (t-statistic = 1.72), indicating that the anomaly is not confined to illiquid small-cap securities where consumption risk effects might be confounded with market microstructure frictions.

Controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the TPB trading strategy maintains an alpha of 35 basis points per month with a t-statistic of 2.84. The signal’s gross Sharpe ratio exceeds 59% of documented anomalies in the factor zoo, while its net Sharpe ratio surpasses 83% of existing strategies. These results establish tax payout burden as an economically significant predictor of cross-sectional returns that captures a distinct dimension of consumption risk exposure not spanned by existing factors or related firm characteristics.

Our paper contributes to the consumption-based asset pricing literature by identifying tax payout burden as a novel firm characteristic that captures exposure to aggregate consumption risk. While [Lettau and Ludvigson \(2001\)](#) demonstrate that consumption-wealth ratios predict returns and [Yogo and Bansal \(2005\)](#) show that durable consumption growth explains cross-sectional variation, we provide direct evidence that firms’ tax-constrained distribution policies create measurable differences in consumption risk exposure. Unlike [Pastor and Veronesi \(2009\)](#), who focus on technological shocks to consumption growth, our mechanism operates through the state-dependent rigidity of tax obligations that prevents firms from optimally

smoothing investor consumption across good and bad states.

Methodologically, we extend the cross-sectional predictability tests in [Fama and French \(2008\)](#) by demonstrating that tax payout burden maintains significant predictive power even after controlling for profitability, investment, and payout yield factors from [Fama and French \(2015\)](#). Our spanning tests reveal that TPB generates monthly alphas of 35 basis points relative to both the six-factor model and the six most closely related anomalies, suggesting the signal captures a unique dimension of systematic risk. The robustness of our results to transaction costs and alternative portfolio constructions addresses concerns raised by [Novy-Marx and Velikov \(2016\)](#) about the implementability of anomaly-based trading strategies.

The broader implications of our findings extend beyond documenting another return predictor to providing new evidence on how operational constraints translate into asset pricing effects through the consumption channel. Tax payout burden emerges as a measurable proxy for firms' inability to provide consumption insurance during high marginal utility states, offering a concrete mechanism linking corporate finance decisions to equilibrium risk premia. Our results suggest that accounting-based measures of payout inflexibility contain important information about firms' fundamental risk characteristics, with tax obligations serving as a particularly powerful signal of consumption risk exposure that commands significant compensation in equilibrium.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S. common stocks traded on major

exchanges, excluding financial firms and utilities due to their unique regulatory environments and capital structures. Tax Payout Burden signal requires two key variables from firms' annual financial statements. Income tax current (TXC) represents the current portion of income tax expense, which is the amount of income taxes payable for the current year based on taxable income. This measure excludes deferred tax expenses and reflects the actual cash tax obligations that firms face in the reporting period. Dividends common (DVC) captures the total cash dividends paid to common shareholders during the fiscal year, representing direct cash distributions to equity holders. We construct the Tax Payout Burden signal by dividing income tax current (TXC) by dividends common (DVC) for each firm-year observation. This ratio captures the relationship between a firm's current tax obligations and its dividend distributions, providing a measure of how tax burdens scale relative to shareholder payouts. We calculate the signal using annual observations at each fiscal year-end, requiring both TXC and DVC to be positive and non-missing. Firms with zero or negative dividends are excluded from the sample as the ratio would be undefined or economically meaningless. The resulting measure indicates the dollar amount of current taxes per dollar of common dividends, with higher values suggesting greater tax obligations relative to dividend distributions.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TPB signal. Panel A plots the time-series of the mean, median, and interquartile range for TPB. On average, the cross-sectional mean (median) TPB is 5.21 (1.37) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input TPB data. The signal's interquartile range spans 0.07 to 4.58. Panel B of Figure 1 plots the time-series of the coverage of the TPB signal for the CRSP universe. On average, the TPB signal

is available for 2.29% of CRSP names, which on average make up 5.31% of total market capitalization.

4 Does TPB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TPB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TPB portfolio and sells the low TPB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short TPB strategy earns an average return of 0.28% per month with a t-statistic of 2.09. The annualized Sharpe ratio of the strategy is 0.28. The alphas range from 0.13% to 0.34% per month and have t-statistics exceeding 0.99 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.52, with a t-statistic of -6.78 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 220 stocks and an average market capitalization of at least \$727 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive

to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.62. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for eight exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 12-46bps/month. The lowest return, (12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.37. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TPB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-four cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the TPB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TPB, as well as average returns and alphas for long/short trading TPB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TPB strategy achieves an average return of 25 bps/month with a t-statistic of 1.72. Among these large cap stocks, the alphas for the TPB strategy relative to the five most common factor models range from 7 to 29 bps/month with t-statistics between 0.51 and 2.31.

5 How does TPB perform relative to the zoo?

Figure 2 puts the performance of TPB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TPB strategy falls in the distribution. The TPB strategy’s gross (net) Sharpe ratio of 0.28 (0.25) is greater than 59% (83%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TPB strategy (red line).² Ignoring trading costs, a \$1 invested in the TPB strategy would have yielded \$3.92 which ranks the TPB strategy in the top 6% across the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the TPB strategy would have yielded \$3.11 which ranks the TPB strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TPB relative to those. Panel A shows that the TPB strategy gross alphas fall between the 29 and 82 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TPB strategy has a positive net generalized alpha for five out of the five factor models. In these cases TPB ranks between the 54 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TPB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TPB with 201 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TPB or at least to weaken the power TPB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TPB conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TPB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TPB}TPB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TPB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TPB. Stocks are finally grouped into five TPB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TPB trading strategies conditioned on each of the 201 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TPB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TPB signal in these Fama-MacBeth regressions exceed 1.41, with the minimum t-statistic occurring when controlling for Long-term EPS forecast. Controlling for all six closely related anomalies, the t-statistic on TPB is

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the TPB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TPB strategy earns alphas that range from 28-38bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.17, which is achieved when controlling for Long-term EPS forecast. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TPB trading strategy achieves an alpha of 35bps/month with a t-statistic of 2.84.

7 Does TPB add relative to the whole zoo?

Finally, we can ask how much adding TPB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the TPB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes TPB grows to \$1963.44.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TPB is available.

8 Conclusion

This study provides compelling evidence that Tax Payout Burden (TPB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TPB-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on TPB delivers an annualized Sharpe ratio of 0.28 gross (0.25 net of transaction costs), indicating strong economic viability even after implementation costs. More importantly, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (33 bps/month gross, 28 bps/month net) suggests that TPB captures a distinct dimension of expected returns not explained by conventional risk factors. The robustness of these results is further validated by the persistence of significant alpha (35 bps/month with t-statistic of 2.84) even after controlling for six closely related strategies, including growth measures, payout characteristics, and asset-based signals.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, TPB emerges as a meaningful addition to the factor zoo that warrants inclusion in asset pricing models and further theoretical investigation. For practitioners, the signal’s robust performance net of transaction costs and its orthogonality to existing factors suggest potential value as either a standalone strategy or as a complement to existing factor portfolios.

However, several limitations merit consideration. First, while our analysis follows state-of-the-art protocols for anomaly testing, the true out-of-sample performance of TPB remains to be observed as the signal gains wider recognition. Second, the economic mechanisms underlying the TPB effect require further theoretical and em-

pirical investigation to determine whether it represents a risk-based explanation or a behavioral anomaly. Third, the stability of the TPB signal across different market regimes and its performance in international markets remain unexplored.

Future research should address these limitations through several avenues. Investigation into the fundamental drivers of the TPB effect could provide insights into whether it reflects systematic risk exposure, limits to arbitrage, or behavioral biases. Cross-country analysis would establish the generalizability of the findings beyond U.S. markets. Additionally, examining the interaction between TPB and corporate events such as tax reform, dividend policy changes, or share repurchase programs could enhance our understanding of the signal's economic foundations. Finally, exploring the optimal combination of TPB with other factors in multi-factor portfolios could yield practical insights for portfolio construction and risk management.

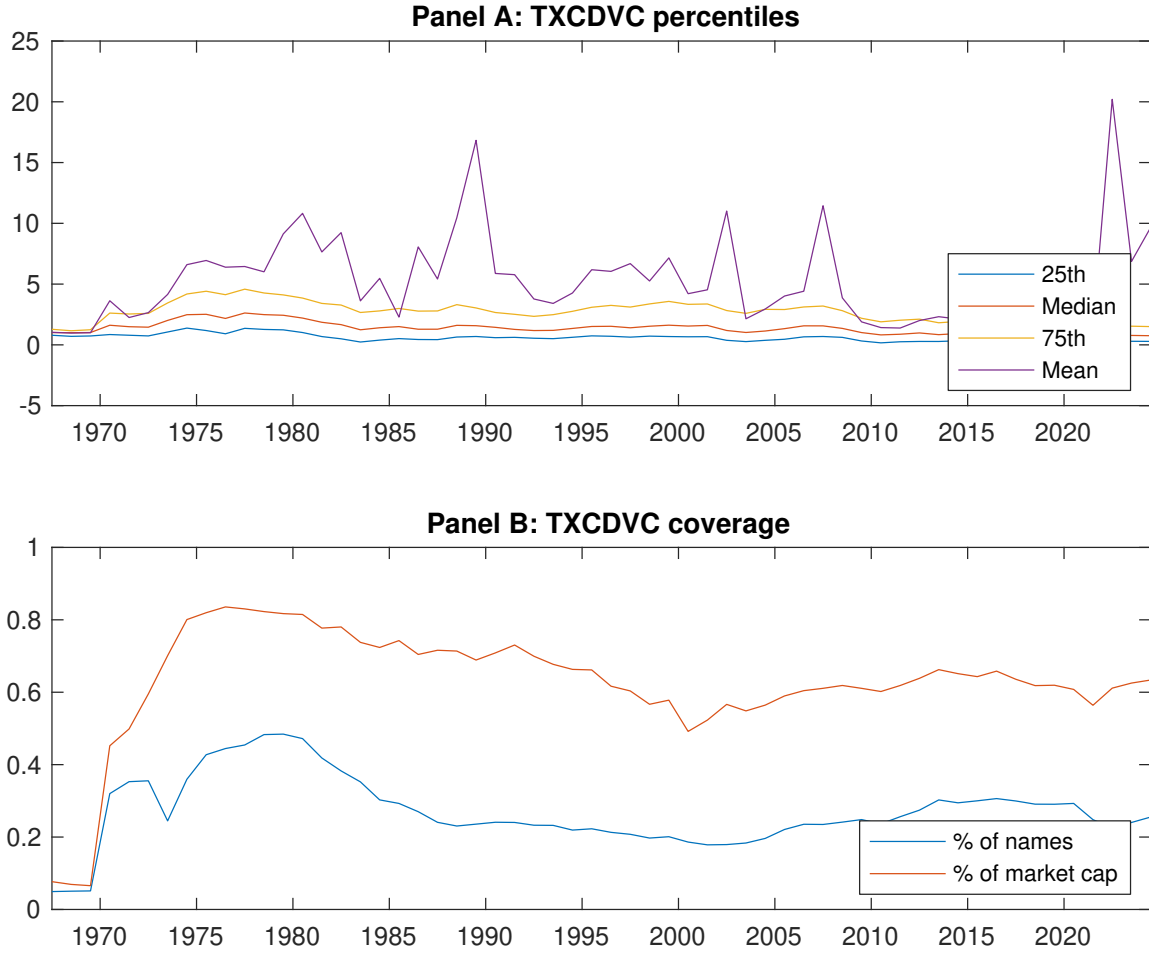


Figure 1: Times series of TPB percentiles and coverage.
This figure plots descriptive statistics for TPB. Panel A shows cross-sectional percentiles of TPB over the sample. Panel B plots the monthly coverage of TPB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TPB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on TPB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.50 [2.99]	0.51 [3.27]	0.58 [3.44]	0.66 [3.61]	0.78 [3.86]	0.28 [2.09]
α_{CAPM}	0.03 [0.29]	0.05 [0.63]	0.07 [0.95]	0.08 [1.33]	0.15 [1.98]	0.13 [0.99]
α_{FF3}	-0.13 [-1.65]	-0.02 [-0.24]	0.02 [0.33]	0.08 [1.33]	0.17 [2.26]	0.30 [2.62]
α_{FF4}	-0.11 [-1.37]	-0.03 [-0.39]	0.04 [0.50]	0.11 [1.78]	0.20 [2.52]	0.30 [2.60]
α_{FF5}	-0.19 [-2.60]	-0.18 [-3.07]	-0.18 [-2.77]	-0.01 [-0.13]	0.14 [1.87]	0.34 [3.03]
α_{FF6}	-0.17 [-2.21]	-0.17 [-2.91]	-0.15 [-2.34]	0.02 [0.39]	0.16 [2.09]	0.33 [2.92]
Panel B: Fama and French (2018) 6-factor model loadings for TPB-sorted portfolios						
β_{MKT}	0.92 [51.80]	0.92 [65.43]	0.96 [63.76]	1.03 [73.24]	1.05 [56.74]	0.13 [4.81]
β_{SMB}	-0.09 [-3.62]	-0.20 [-9.67]	-0.08 [-3.45]	-0.07 [-3.34]	0.10 [3.73]	0.19 [4.99]
β_{HML}	0.24 [7.02]	0.01 [0.54]	-0.00 [-0.09]	-0.07 [-2.63]	-0.04 [-1.14]	-0.28 [-5.46]
β_{RMW}	-0.05 [-1.58]	0.21 [7.76]	0.40 [13.53]	0.17 [6.03]	0.17 [4.73]	0.23 [4.33]
β_{CMA}	0.41 [8.05]	0.40 [9.95]	0.29 [6.65]	0.15 [3.80]	-0.11 [-2.05]	-0.52 [-6.78]
β_{UMD}	-0.04 [-2.33]	-0.01 [-0.80]	-0.04 [-2.60]	-0.05 [-3.32]	-0.03 [-1.56]	0.01 [0.47]
Panel C: Average number of firms (n) and market capitalization (me)						
n	246	220	228	251	311	
me (\$10 ⁶)	727	1667	1725	1654	1075	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TPB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [2.09]	0.13 [0.99]	0.30 [2.62]	0.30 [2.60]	0.34 [3.03]	0.33 [2.92]
Quintile	NYSE	EW	0.22 [2.62]	0.13 [1.58]	0.19 [2.47]	0.18 [2.28]	0.11 [1.61]	0.11 [1.46]
Quintile	Name	VW	0.31 [2.24]	0.15 [1.15]	0.33 [2.81]	0.34 [2.82]	0.37 [3.21]	0.37 [3.13]
Quintile	Cap	VW	0.26 [2.03]	0.09 [0.75]	0.24 [2.23]	0.27 [2.46]	0.31 [2.93]	0.32 [3.01]
Decile	NYSE	VW	0.50 [3.17]	0.36 [2.34]	0.54 [3.80]	0.56 [3.85]	0.58 [4.12]	0.59 [4.09]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [1.90]	0.11 [0.89]	0.27 [2.34]	0.27 [2.37]	0.29 [2.59]	0.28 [2.49]
Quintile	NYSE	EW	0.12 [1.37]	0.03 [0.40]	0.09 [1.12]	0.09 [1.07]	0.02 [0.11]	
Quintile	Name	VW	0.29 [2.05]	0.14 [1.05]	0.30 [2.52]	0.31 [2.57]	0.32 [2.75]	0.31 [2.68]
Quintile	Cap	VW	0.23 [1.85]	0.08 [0.66]	0.21 [2.00]	0.23 [2.18]	0.26 [2.47]	0.26 [2.49]
Decile	NYSE	VW	0.46 [2.93]	0.34 [2.18]	0.50 [3.49]	0.51 [3.55]	0.52 [3.66]	0.51 [3.63]

Table 3: Conditional sort on size and TPB

This table presents results for conditional double sorts on size and TPB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TPB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TPB and short stocks with low TPB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TPB Quintiles					TPB Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.67 [2.89]	0.96 [4.02]	0.83 [4.06]	1.28 [3.46]	0.99 [4.06]	0.32 [2.28]	0.25 [1.76]	0.30 [2.12]	0.23 [1.65]	0.26 [1.81]	0.20 [1.43]
	(2)	0.78 [3.71]	0.79 [4.04]	0.76 [3.67]	0.84 [3.84]	0.81 [3.44]	0.03 [0.27]	-0.05 [-0.39]	0.00 [0.01]	0.02 [0.19]	-0.09 [-0.78]	-0.07 [-0.61]
	(3)	0.68 [3.51]	0.65 [3.61]	0.83 [4.29]	0.87 [4.12]	0.86 [3.81]	0.19 [1.49]	0.07 [0.59]	0.16 [1.37]	0.18 [1.47]	0.05 [0.48]	0.07 [0.60]
	(4)	0.62 [3.34]	0.69 [3.88]	0.71 [3.76]	0.68 [3.54]	0.77 [3.42]	0.15 [1.13]	-0.01 [-0.08]	0.10 [0.83]	0.13 [1.09]	0.02 [0.20]	0.05 [0.43]
	(5)	0.49 [3.16]	0.48 [2.98]	0.54 [3.13]	0.60 [3.31]	0.74 [3.62]	0.25 [1.72]	0.07 [0.51]	0.21 [1.66]	0.21 [1.63]	0.29 [2.31]	0.28 [2.17]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TPB Quintiles					TPB Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	85	86	86	86	86	6	7	7	8	8	
	(2)	44	44	44	44	44	19	20	20	20	20	
	(3)	39	39	39	39	39	44	44	45	45	44	
	(4)	39	39	39	39	39	119	121	119	119	117	
(5)	43	43	43	43	43	922	1261	1221	1446	1046		

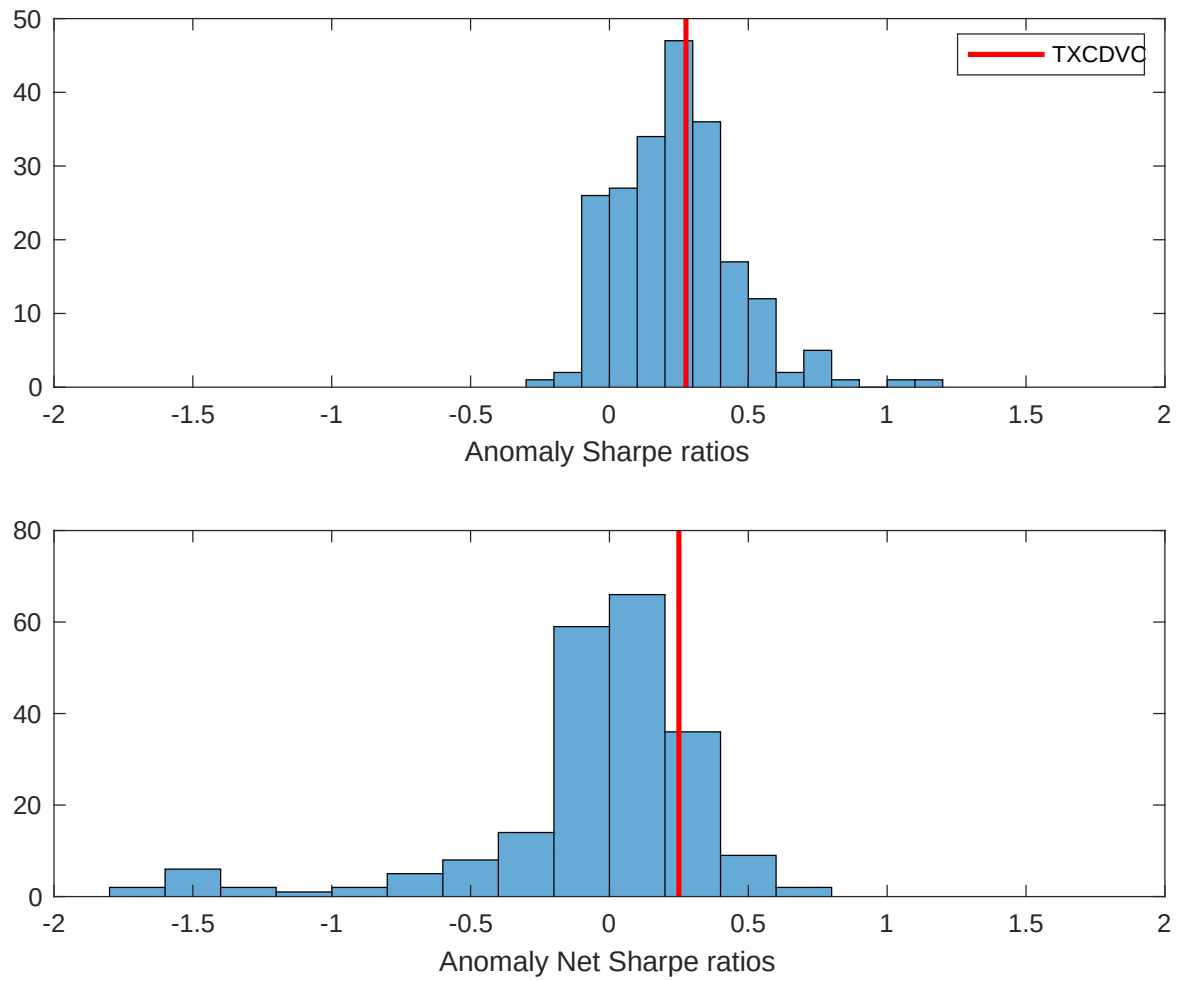


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TPB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

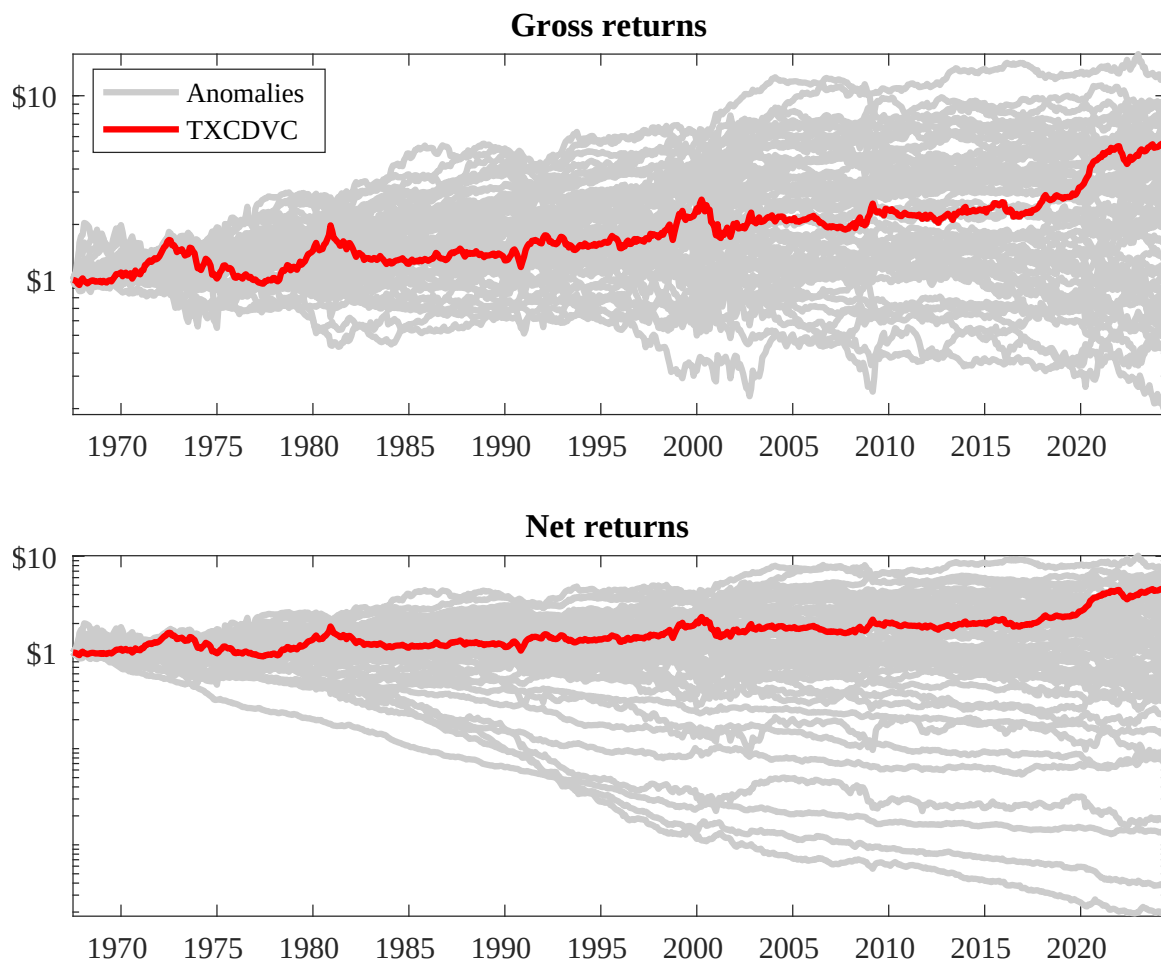
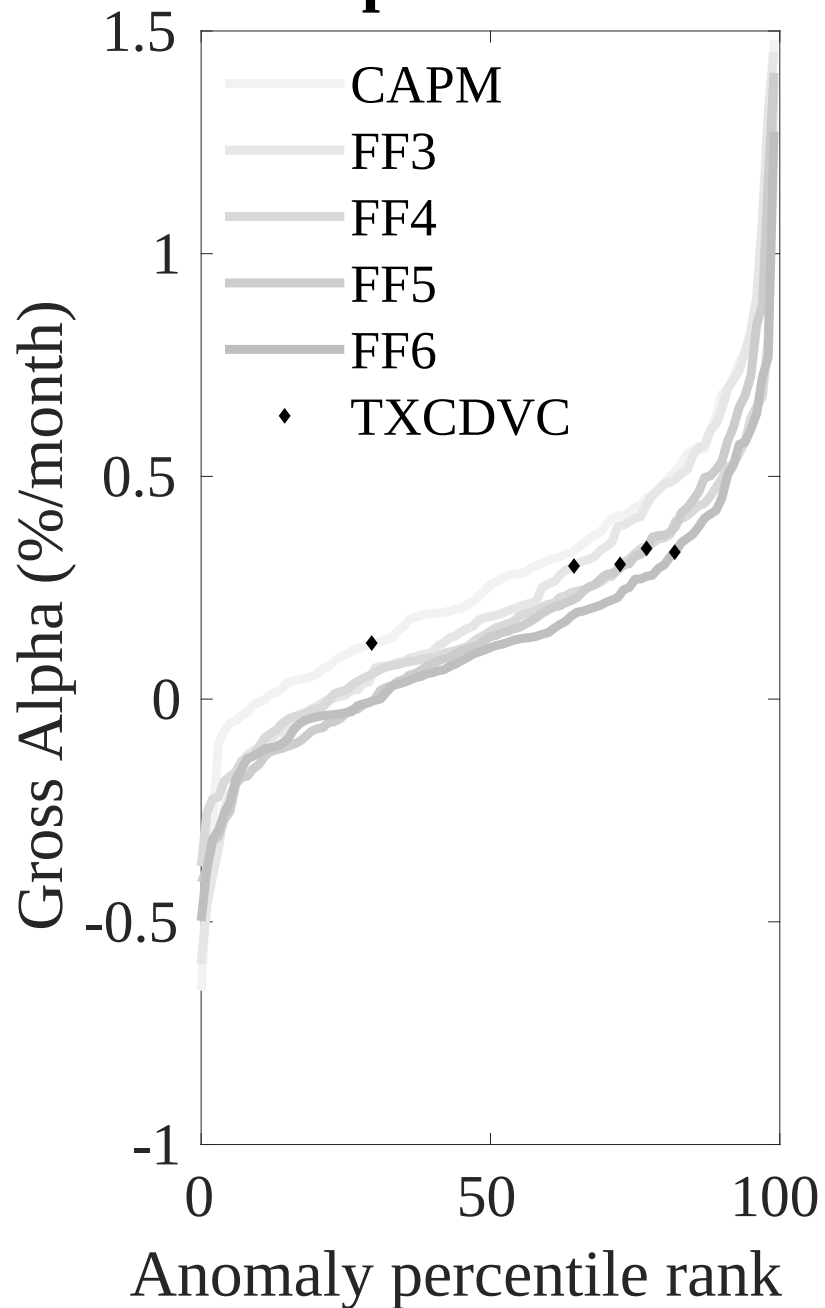


Figure 3: Dollar invested.

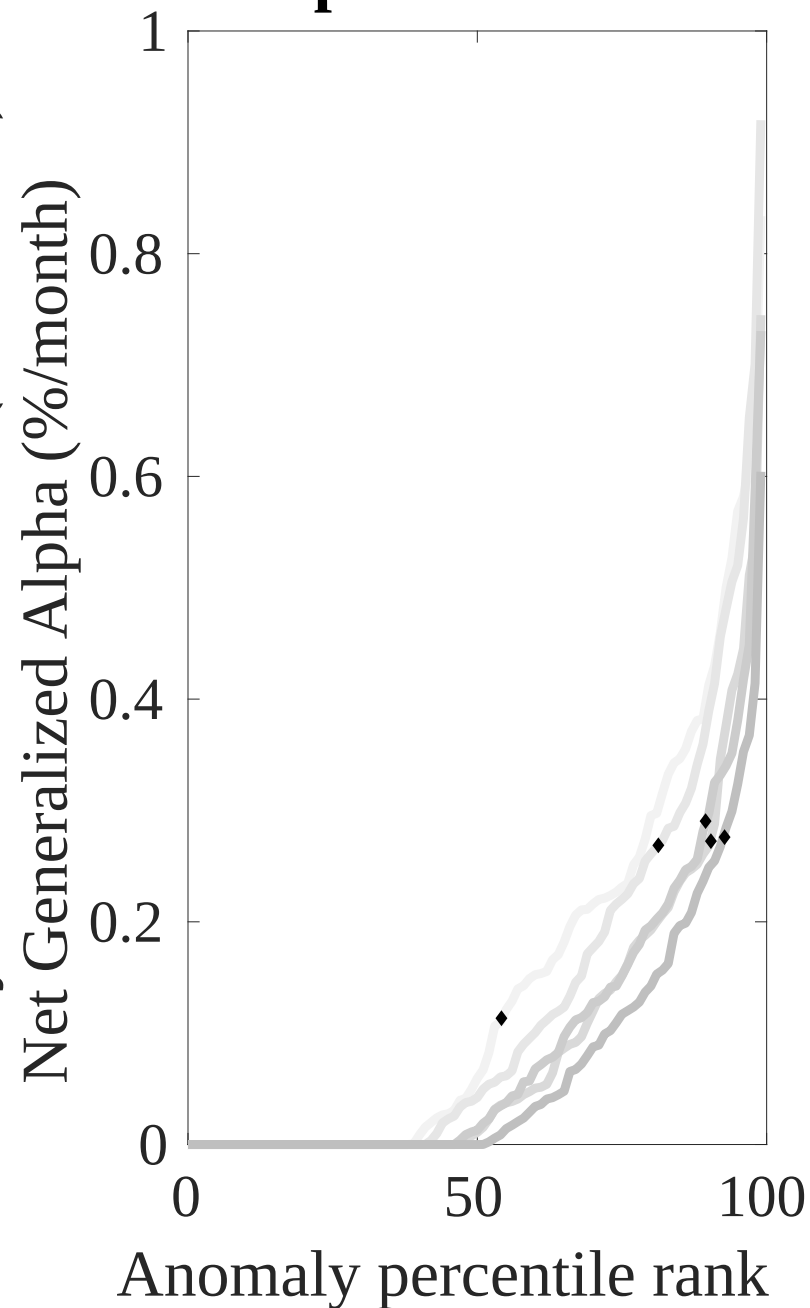
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TPB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



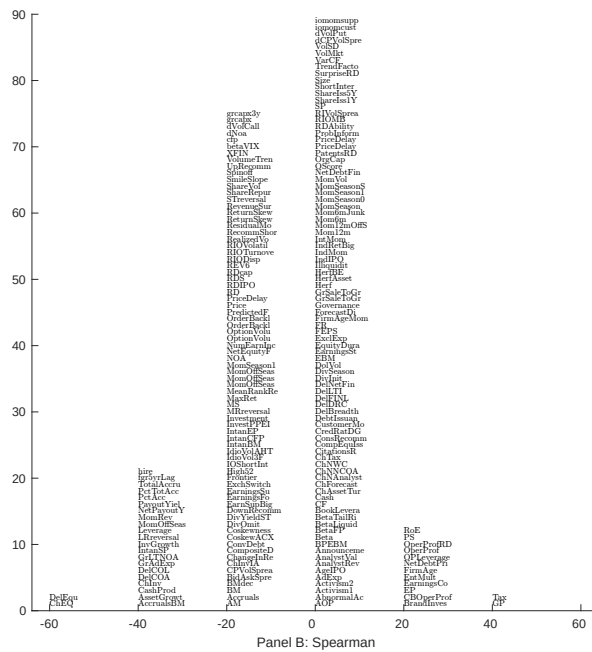
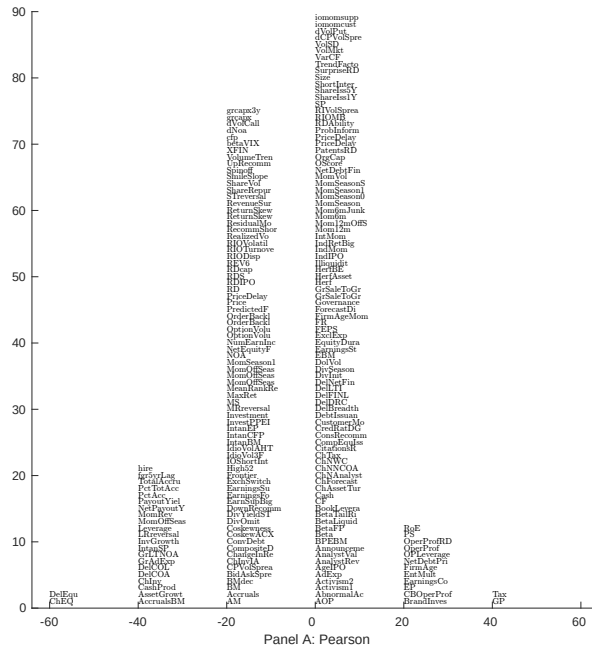


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 201 filtered anomaly signals with TPB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

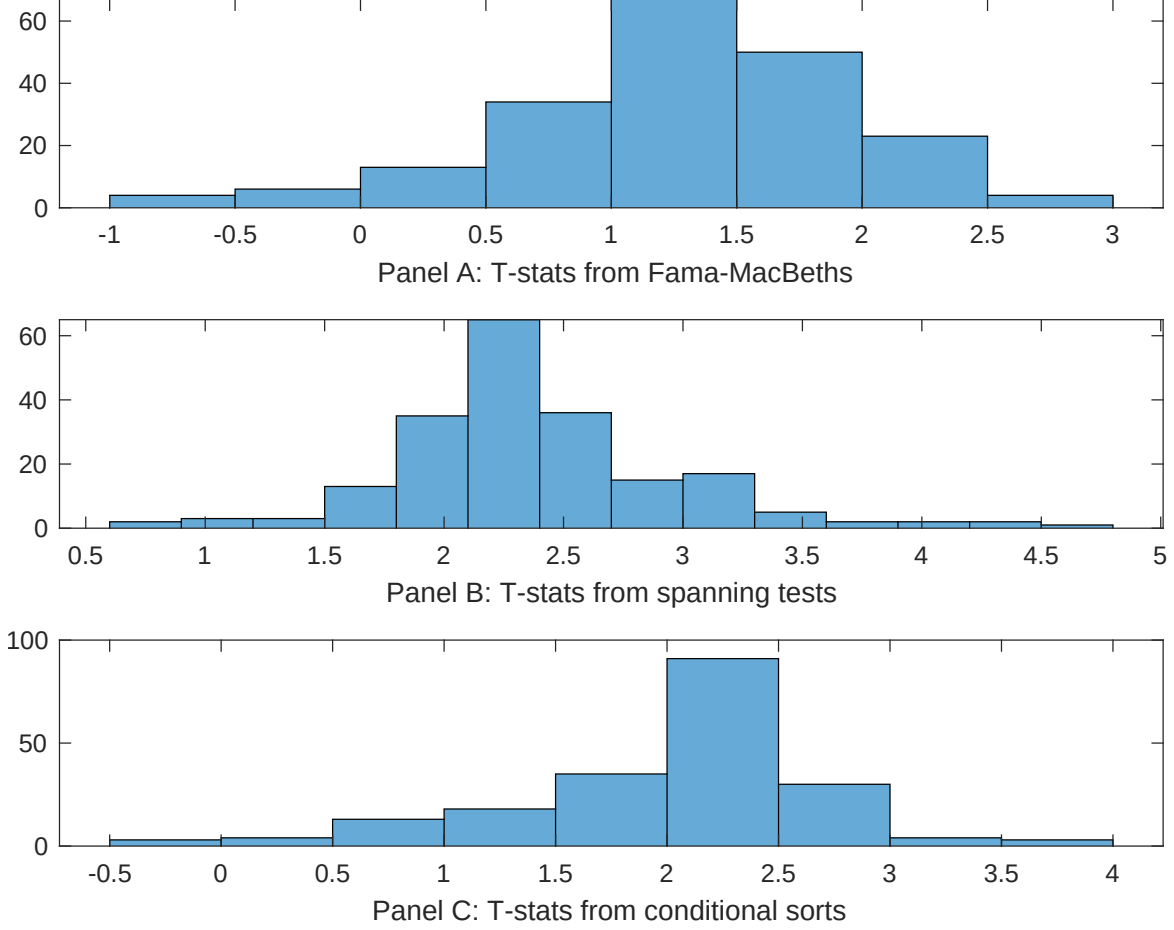


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TPB conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TPB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TPB}TPB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TPB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TPB. Stocks are finally grouped into five TPB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TPB trading strategies conditioned on each of the 201 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TPB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TPB}TPB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Long-term EPS forecast, Payout Yield, Asset growth, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.18 [6.61]	0.12 [6.50]	0.13 [7.12]	0.12 [6.21]	0.12 [6.85]	0.12 [6.53]	0.14 [6.76]
TPB	0.29 [2.10]	0.32 [2.27]	0.14 [1.41]	0.27 [2.14]	0.28 [1.97]	0.26 [1.85]	0.21 [1.88]
Anomaly 1	0.59 [2.97]						0.15 [1.08]
Anomaly 2		0.16 [2.50]					-0.29 [-0.53]
Anomaly 3			0.95 [1.26]				0.40 [0.50]
Anomaly 4				0.20 [0.45]			-0.79 [-0.12]
Anomaly 5					0.84 [5.88]		0.38 [2.77]
Anomaly 6						0.13 [3.30]	0.92 [0.17]
# months	684	684	504	668	684	684	499
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TPB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TPB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Long-term EPS forecast, Payout Yield, Asset growth, Change in current operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.36 [3.37]	0.34 [3.14]	0.28 [2.17]	0.38 [3.49]	0.34 [3.08]	0.34 [3.03]	0.35 [2.84]
Anomaly 1	-47.18 [-7.90]						-24.25 [-2.17]
Anomaly 2		-42.69 [-7.87]					-9.22 [-0.80]
Anomaly 3			-35.05 [-8.12]				-25.23 [-5.32]
Anomaly 4				-30.71 [-7.91]			-10.45 [-1.93]
Anomaly 5					-40.99 [-6.28]		-7.83 [-0.87]
Anomaly 6						-9.43 [-1.71]	6.80 [1.07]
mkt	11.40 [4.47]	13.70 [5.38]	-1.19 [-0.36]	7.70 [2.94]	12.92 [5.00]	12.99 [4.89]	-1.25 [-0.38]
smb	21.72 [5.79]	20.75 [5.54]	8.56 [1.82]	17.37 [4.64]	23.82 [6.20]	18.47 [4.53]	10.98 [2.26]
hml	-22.05 [-4.41]	-21.81 [-4.35]	-3.50 [-0.58]	-12.25 [-2.31]	-25.68 [-5.09]	-23.66 [-4.22]	-4.81 [-0.76]
rmw	20.46 [4.10]	18.64 [3.72]	24.70 [4.17]	20.82 [4.19]	23.41 [4.62]	21.60 [4.12]	22.67 [3.86]
cma	-6.64 [-0.70]	-9.65 [-1.05]	-32.35 [-3.57]	-39.32 [-5.19]	-4.10 [-0.38]	-48.26 [-5.87]	2.78 [0.23]
umd	2.27 [0.89]	0.45 [0.18]	2.81 [0.97]	-3.71 [-1.43]	0.32 [0.13]	0.95 [0.36]	1.40 [0.47]
# months	684	684	504	680	684	684	500
$\bar{R}^2(\%)$	43	43	41	43	41	38	46

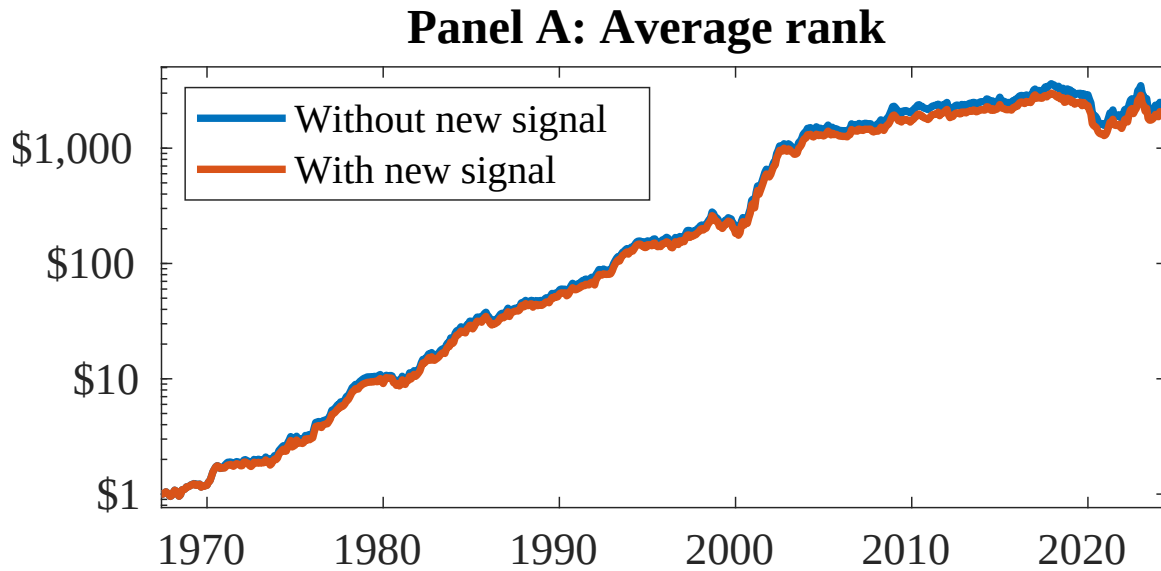


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as TPB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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